

SOLIDWORKS API (NET VBA C#) — Reference (263/337): ISketch (part 1 of 3)

SOLIDWORKS API Documentation (converted from CHM)

ISketch (part 1 of 3) (Type)

SOLIDWORKS API Help

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ISketch Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketch Interface

Allows access to sketch entities and to extract information about sketch elements, the sketch orientation, and so on.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISketch

Visual Basic (Usage)



Dim instance As [ISketch](#)

C#

public interface ISketch

C++/CLI

public interface class ISketch

Visual Basic for Applications (VBA) Syntax

See [Sketch](#).

Example

[Get Spline Points \(C++\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)
[Create Imported Surface Body from Sketch \(C#\)](#)
[Delete All Constraints in Selected Sketch \(VBA\)](#)
[Transform Sketch to Model \(VBA\)](#)
[Open and Exit Sketch \(VBA\)](#)

Remarks

A sketch is typically used in [part](#) or [assembly](#) documents to generate a solid body or such subsequent [features](#) as a cut or boss. A sketch is also used in [drawing](#) documents whenever the user constructs individual lines, arcs, and so on, on a drawing [sheet](#) or in a drawing [view](#).

Accessors

[IBoundaryBossFeatureData::D1Curves](#)
[IBoundaryBossFeatureData::D2Curves](#)
[IChainPatternFeatureData::Path](#)
[ICosmeticWeldBeadFeatureData::GetEntitiesWeldPath](#)
[ICurveDrivenPatternFeatureData::D1Direction](#)
[ICurveDrivenPatternFeatureData::D2Direction](#)
[IDomeFeatureData2::ConstraintPointOrSketch](#)
[IFeature::GetSpecificFeature2](#)
[ILocalCurvePatternFeatureData::D1Direction](#)
[ILocalCurvePatternFeatureData::D2Direction](#)
[ILocalSketchPatternFeatureData::Sketch](#)
[IModelDoc2::GetActiveSketch2](#) and [IModelDoc2::!GetActiveSketch2](#)

[INote::GetSketch](#) or [INote::IGetSketch](#)
[IProfileGroupFolder::GetSketch](#)
[ISheet::GetTemplateSketch](#)
[ISketchBlockDefinition::GetSketch](#)
[ISketchBlockInstance::GetSketch](#)
[ISketchContour::Sketch](#)
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[IStructureSystemMemberProfile::GetSketch](#)
[ISweptFlangeFeatureData::Path](#)
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[IView::GetCenterLineSketch](#)
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[IWrapSketchFeatureData::GetSourceSketch](#)

 Access Diagram

[Sketch](#)

 See Also

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[ISketchText Interface](#)

ISketch Interface Methods

[See Also](#)


SolidWorks.Interop.sldworks Namespace : ISketch Interface

For a list of all members of this type, see [ISketch members](#).

 Public Methods

Name	Description
 AutoDimension	Obsolete. Superseded by ISketch.AutoDimension2
 AutoDimension2	Obsolete. Superseded by ISketchManager.FullyDefineSketch .
 CheckFeatureUse	Checks to see if this sketch is valid for use in creating a specified feature.
 ConstrainAll	Attempts to solve all of the apparent relations in the sketch and returns the number of constraints that were added to the sketch.
 GetArcCount	Gets the number of arcs in the sketch.
 GetArcs	Obsolete. Superseded by ISketch.GetArcs2 and ISketch.IGetArcs2

	GetArcs2	Gets all of the arcs in the sketch.
	GetAutomaticSolve	Checks whether the computation to solve the sketch geometry of the part as modifications are automatically performed.
	GetConstrainedStatus	Gets the current constrained status of the sketch.
	GetContourEdgeCount	Gets the number of edges for this sketch contour.
	GetContourEdges	Gets the edges for a sketch that has one contour.
	GetDetachSegmentOnDrag	Gets the Detach Segment on Drag setting.
	GetEllipseCount	Gets the number of ellipses in the sketch.
	GetEllipses	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3 .
	GetEllipses2	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3 .
	GetEllipses3	Gets all of the ellipses in the sketch.
	GetLineCount	Obsolete. Superseded by ISketch::GetLineCount2 .
	GetLineCount2	Gets the number of lines in the sketch with an option to exclude or include crosshatch lines.
	GetLines	Obsolete. Superseded by ISketch::GetLines2 and ISketch::IGetLines2 .
	GetLines2	Gets all of the lines in the sketch with an option to include or exclude crosshatch lines.
	GetParabolaCount	Gets the number of parabolas in the sketch.
	GetParabolas	Obsolete. Superseded by ISketch::GetParabolas2 and ISketch::IGetParabolas2 .
	GetParabolas2	Gets all of the parabolas in the sketch.
	GetPolyLineCount	Not implemented.
	GetPolylines	Not implemented.
	GetReferenceEntity	Gets the entity on which this sketch was created.
	GetSketchBlockInstanceCount	Gets the number of block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).
	GetSketchBlockInstances	Gets the block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).
	GetSketchContourCount	Gets the number of sketch contours in this sketch.
	GetSketchContours	Gets the sketch contours in this sketch.
	GetSketchHatches	Gets an array of sketch hatches that exist in this sketch.
	GetSketchPathCount	Gets the number of sketch paths in this sketch.
	GetSketchPaths	Gets the sketch paths in this sketch.
	GetSketchPictureCount	Gets the number of pictures on this sketch.
	GetSketchPictures	Gets the pictures on this sketch.
	GetSketchPoints	Obsolete. Superseded by ISketch::GetSketchPoints2 , ISketch::IGetSketchPoints2 , and ISketch::IEnumSketchPoints .
	GetSketchPoints2	Gets the sketch points in this sketch.
	GetSketchPointsCount	Obsolete. Superseded by ISketch::GetSketchPointsCount2 .
	GetSketchPointsCount2	Gets the number of sketch points in this sketch.
	GetSketchRegionCount	Gets the number of sketch regions in this sketch.
	GetSketchRegions	Gets the sketch regions in this sketch.
	GetSketchSegments	Gets the sketch segments in this sketch, which include line, arc, spline, parabola, and ellipse entities.
	GetSketchSlotCount	Gets the number of sketch slots in this sketch.
	GetSketchSlots	Gets the sketch slots in this sketch.
	GetSketchTextSegments	Gets the sketch segments that represent the selected text in the sketch.
	GetSplineCount	Gets the number of splines in this sketch.
	GetSplineInterpolateCount	Gets the number of points in the spline and number of splines in the sketch.
	GetSplineParams	Obsolete. Superseded by ISketch::GetSplineParams2 and ISketch::IGetSplineParams2 .
	GetSplineParams2	Obsolete. Superseded by ISketch::GetSplineParams3 .

		
	GetSplineParams3	Obsolete. Superseded by ISketch::GetSplineParams4
	GetSplineParams4	Obsolete. Superseded by ISketch::GetSplineParams5
	GetSplineParams5	Gets the parameterization data of the specified spline in this sketch.
	GetSplineParamsCount	Obsolete. Superseded by ISketch::GetSplineParamsCount2
	GetSplineParamsCount2	Obsolete. Superseded by ISketch::GetSplineParamsCount3
	GetSplineParamsCount3	Gets the number of splines in the sketch and the size of array required to hold the data for them.
	GetSplines	Gets information for each spline by tessellation instead of by interpolation as is done by ISketch::GetSplinesInterpolate and ISketch::IGetSplinesInterpolate .
	GetSplinesInterpolate	Gets the spline points by interpolation instead of by tessellation as is done by ISketch::GetSplines and ISketch::IGetSplines .
	GetUserPoints	Obsolete. Superseded by ISketch::GetUserPoints2 and ISketch::IGetUserPoints2
	GetUserPoints2	Gets all of the user points in this sketch.
	GetUserPointsCount	Gets the number of user points in the sketch.
	IEnumSketchHatches	Gets the sketch hatches enumeration in this sketch.
	IEnumSketchPoints	Gets the sketch points enumeration in this sketch.
	IEnumSketchSegments	Gets the sketch segments enumeration in this sketch.
	IEnumSketchTextSegments	Gets the sketch segments enumeration for the selected text in this sketch.
	IGetArcs	Obsolete. Superseded by ISketch::GetArcs2 and ISketch::IGetArcs2
	IGetArcs2	Gets all of the arcs in the sketch.
	IGetContourEdges	Gets the edges for a sketch that has one contour.
	IGetEllipses	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3
	IGetEllipses2	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3
	IGetEllipses3	Gets all of the ellipses in the sketch.
	IGetLines	Obsolete. Superseded by ISketch::GetLines2 and ISketch::IGetLines2
	IGetLines2	Gets all of the lines in the sketch with an option to include or exclude crosshatch lines.
	IGetParabolas	Obsolete. Superseded by ISketch::GetParabolas2 and ISketch::IGetParabolas2
	IGetParabolas2	Gets all of the parabolas in the sketch.
	IGetPolylines	Not implemented.
	IGetSketchBlockInstances	Gets the block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).
	IGetSketchContours	Gets the sketch contours in this sketch.
	IGetSketchPaths	Gets the sketch paths in this sketch.
	IGetSketchPictures	Gets the pictures on this sketch.
	IGetSketchPoints2	Gets the sketch points in this sketch.
	IGetSketchRegions	Gets the sketch regions in this sketch.
	IGetSketchSlots	Gets the sketch slots in this sketch.
	IGetSplineParams	Obsolete. Superseded by ISketch::GetSplineParams2 and ISketch::IGetSplineParams2
	IGetSplineParams2	Obsolete. Superseded by ISketch::IGetSplineParameters3
	IGetSplineParams3	Obsolete. Superseded by ISketch::GetSplineParams4
	IGetSplines	Gets information for each spline by tessellation instead of by interpolation as is done by ISketch::GetSplinesInterpolate and ISketch::IGetSplinesInterpolate .
	IGetSplinesInterpolate	Gets the spline points by interpolation instead of by tessellation as is done by ISketch::GetSplines and ISketch::IGetSplines .
	IGetUserPoints	Obsolete. Superseded by ISketch::GetUserPoints2 and ISketch::IGetUserPoints2
	IGetUserPoints2	Gets all of the user points in this sketch.
	InsertRouteLine	Inserts a route line in an explode line sketch or a 3D sketch to indicate component relationships.
	Is3D	Gets whether this sketch is 2D or 3D.

		
	IsBoundingBoxSketch	Determines whether the sketch is a boundary box.
	IsDerived	Gets whether a sketch is derived.
	IsShared	Gets whether this sketch is used by more than one feature.
	IsSketchEditable	Gets whether this sketch is editable.
	MergePoints	Merges sketch points within a specified distance.
	SetAutomaticSolve	Controls whether the computation to solve the sketch geometry of the part as modifications are automatically performed.
	SetDetachSegmentOnDrag	Sets the Detach Segment on Drag setting.
	SetSketchEditable	Sets whether this sketch is editable.
	SetWorkingPlaneOrientation	Sets the orientation for sketching geometry in a 3D sketch. It sets the planar location for new 2D and 3D geometry in a 3D sketch.

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ISketch Interface Members

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SolidWorks.Interop.sldworks Namespace	: ISketch Interface

The following tables list the members exposed by [ISketch](#).

Public Properties

Name	Description
 IModelToSketchXform	Obsolete. Superseded by ISketch::ModelToSketchTransform. NOTE: This property is a get-only property. Set is not implemented.
 ModelToSketchTransform	Gets the model-to-sketch transform for this sketch. NOTE: This property is a get-only property. Set is not implemented.
 ModelToSketchXform	Obsolete. Superseded by ISketch::ModelToSketchTransform. NOTE: This property is a get-only property. Set is not implemented.
 RelationManager	Gets the sketch relation manager. NOTE: This property is a get-only property. Set is not implemented.

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Public Methods

Name	Description
 AutoDimension	Obsolete. Superseded by ISketch::AutoDimension2.
 AutoDimension2	Obsolete. Superseded by ISketchManager::FullyDefineSketch.

	CheckFeatureUse	Checks to see if this sketch is valid for use in creating a specified feature.
	ConstrainAll	Attempts to solve all of the apparent relations in the sketch and returns the number of constraints that were added to the sketch.
	GetArcCount	Gets the number of arcs in the sketch.
	GetArcs	Obsolete. Superseded by ISketch::GetArcs2 and ISketch::IGetArcs2 .
	GetArcs2	Gets all of the arcs in the sketch.
	GetAutomaticSolve	Checks whether the computation to solve the sketch geometry of the part as modifications are automatically performed.
	GetConstrainedStatus	Gets the current constrained status of the sketch.
	GetContourEdgeCount	Gets the number of edges for this sketch contour.
	GetContourEdges	Gets the edges for a sketch that has one contour.
	GetDetachSegmentOnDrag	Gets the Detach Segment on Drag setting.
	GetEllipseCount	Gets the number of ellipses in the sketch.
	GetEllipses	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3 .
	GetEllipses2	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3 .
	GetEllipses3	Gets all of the ellipses in the sketch.
	GetLineCount	Obsolete. Superseded by ISketch::GetLineCount2 .
	GetLineCount2	Gets the number of lines in the sketch with an option to exclude or include crosshatch lines.
	GetLines	Obsolete. Superseded by ISketch::GetLines2 and ISketch::IGetLines2 .
	GetLines2	Gets all of the lines in the sketch with an option to include or exclude crosshatch lines.
	GetParabolaCount	Gets the number of parabolas in the sketch.
	GetParabolas	Obsolete. Superseded by ISketch::GetParabolas2 and ISketch::IGetParabolas2 .
	GetParabolas2	Gets all of the parabolas in the sketch.
	GetPolyLineCount	Not implemented.
	GetPolylines	Not implemented.
	GetReferenceEntity	Gets the entity on which this sketch was created.
	GetSketchBlockInstanceCount	Gets the number of block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).
	GetSketchBlockInstances	Gets the block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).
	GetSketchContourCount	Gets the number of sketch contours in this sketch.
	GetSketchContours	Gets the sketch contours in this sketch.
	GetSketchHatches	Gets an array of sketch hatches that exist in this sketch.
	GetSketchPathCount	Gets the number of sketch paths in this sketch.
	GetSketchPaths	Gets the sketch paths in this sketch.
	GetSketchPictureCount	Gets the number of pictures on this sketch.
	GetSketchPictures	Gets the pictures on this sketch.
	GetSketchPoints	Obsolete. Superseded by ISketch::GetSketchPoints2 , ISketch::IGetSketchPoints2 , and ISketch::IEnumSketchPoints .
	GetSketchPoints2	Gets the sketch points in this sketch.
	GetSketchPointsCount	Obsolete. Superseded by ISketch::GetSketchPointsCount2 .
	GetSketchPointsCount2	Gets the number of sketch points in this sketch.
	GetSketchRegionCount	Gets the number of sketch regions in this sketch.
	GetSketchRegions	Gets the sketch regions in this sketch.
	GetSketchSegments	Gets the sketch segments in this sketch, which include line, arc, spline, parabola, and ellipse entities.
	GetSketchSlotCount	Gets the number of sketch slots in this sketch.
	GetSketchSlots	Gets the sketch slots in this sketch.
	GetSketchTextSegments	Gets the sketch segments that represent the selected text in the sketch.

		
	GetSplineCount	Gets the number of splines in this sketch.
	GetSplineInterpolateCount	Gets the number of points in the spline and number of splines in the sketch.
	GetSplineParams	Obsolete. Superseded by ISketch::GetSplineParams2 and ISketch::IGetSplineParams2 .
	GetSplineParams2	Obsolete. Superseded by ISketch::GetSplineParams3 .
	GetSplineParams3	Obsolete. Superseded by ISketch::GetSplineParams4 .
	GetSplineParams4	Obsolete. Superseded by ISketch::GetSplineParams5 .
	GetSplineParams5	Gets the parameterization data of the specified spline in this sketch.
	GetSplineParamsCount	Obsolete. Superseded by ISketch::GetSplineParamsCount2 .
	GetSplineParamsCount2	Obsolete. Superseded by ISketch::GetSplineParamsCount3 .
	GetSplineParamsCount3	Gets the number of splines in the sketch and the size of array required to hold the data for them.
	GetSplines	Gets information for each spline by tessellation instead of by interpolation as is done by ISketch::GetSplinesInterpolate and ISketch::IGetSplinesInterpolate .
	GetSplinesInterpolate	Gets the spline points by interpolation instead of by tessellation as is done by ISketch::GetSplines and ISketch::IGetSplines .
	GetUserPoints	Obsolete. Superseded by ISketch::GetUserPoints2 and ISketch::IGetUserPoints2 .
	GetUserPoints2	Gets all of the user points in this sketch.
	GetUserPointsCount	Gets the number of user points in the sketch.
	IEnumSketchHatches	Gets the sketch hatches enumeration in this sketch.
	IEnumSketchPoints	Gets the sketch points enumeration in this sketch.
	IEnumSketchSegments	Gets the sketch segments enumeration in this sketch.
	IEnumSketchTextSegments	Gets the sketch segments enumeration for the selected text in this sketch.
	IGetArcs	Obsolete. Superseded by ISketch::GetArcs2 and ISketch::IGetArcs2 .
	IGetArcs2	Gets all of the arcs in the sketch.
	IGetContourEdges	Gets the edges for a sketch that has one contour.
	IGetEllipses	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3 .
	IGetEllipses2	Obsolete. Superseded by ISketch::GetEllipses3 and ISketch::IGetEllipses3 .
	IGetEllipses3	Gets all of the ellipses in the sketch.
	IGetLines	Obsolete. Superseded by ISketch::GetLines2 and ISketch::IGetLines2 .
	IGetLines2	Gets all of the lines in the sketch with an option to include or exclude crosshatch lines.
	IGetParabolas	Obsolete. Superseded by ISketch::GetParabolas2 and ISketch::IGetParabolas2 .
	IGetParabolas2	Gets all of the parabolas in the sketch.
	IGetPolyLines	Not implemented.
	IGetSketchBlockInstances	Gets the block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).
	IGetSketchContours	Gets the sketch contours in this sketch.
	IGetSketchPaths	Gets the sketch paths in this sketch.
	IGetSketchPictures	Gets the pictures on this sketch.
	IGetSketchPoints2	Gets the sketch points in this sketch.
	IGetSketchRegions	Gets the sketch regions in this sketch.
	IGetSketchSlots	Gets the sketch slots in this sketch.
	IGetSplineParams	Obsolete. Superseded by ISketch::GetSplineParams2 and ISketch::IGetSplineParams2 .
	IGetSplineParams2	Obsolete. Superseded by ISketch::IGetSplineParameters3 .
	IGetSplineParams3	Obsolete. Superseded by ISketch::GetSplineParams4 .
	IGetSplines	Gets information for each spline by tessellation instead of by interpolation as is done by ISketch::GetSplinesInterpolate and ISketch::IGetSplinesInterpolate .
	IGetSplinesInterpolate	Gets the spline points by interpolation instead of by tessellation as is done by ISketch::GetSplines and ISketch::IGetSplines .

		
	IGetUserPoints	Obsolete. Superseded by ISketch.GetUserPoints2 and ISketch.IGetUserPoints2
	IGetUserPoints2	Gets all of the user points in this sketch.
	InsertRouteLine	Inserts a route line in an explode line sketch or a 3D sketch to indicate component relationships.
	Is3D	Gets whether this sketch is 2D or 3D.
	IsBoundaryBoxSketch	Determines whether the sketch is a boundary box.
	IsDerived	Gets whether a sketch is derived.
	IsShared	Gets whether this sketch is used by more than one feature.
	IsSketchEditable	Gets whether this sketch is editable.
	MergePoints	Merges sketch points within a specified distance.
	SetAutomaticSolve	Controls whether the computation to solve the sketch geometry of the part as modifications are automatically performed.
	SetDetachSegmentOnDrag	Sets the Detach Segment on Drag setting.
	SetSketchEditable	Sets whether this sketch is editable.
	SetWorkingPlaneOrientation	Sets the orientation for sketching geometry in a 3D sketch. It sets the planar location for new 2D and 3D geometry in a 3D sketch.

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ISketch Interface Properties

[See Also](#)



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For a list of all members of this type, see [ISketch members](#).

Public Properties

Name	Description
 IModelToSketchXform	Obsolete. Superseded by ISketch::ModelToSketchTransform. NOTE: This property is a get-only property. Set is not implemented.
 ModelToSketchTransform	Gets the model-to-sketch transform for this sketch. NOTE: This property is a get-only property. Set is not implemented.
 ModelToSketchXform	Obsolete. Superseded by ISketch::ModelToSketchTransform. NOTE: This property is a get-only property. Set is not implemented.
 RelationManager	Gets the sketch relation manager. NOTE: This property is a get-only property. Set is not implemented.

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AutoDimension Method (ISketch)

[See Also](#)



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Obsolete. Superseded by [ISketch::AutoDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AutoDimension( _  
    ByVal EntitiesToDimension As System.Integer, _  
    ByVal HorizontalScheme As System.Integer, _  
    ByVal HorizontalPlacement As System.Integer, _  
    ByVal VerticalScheme As System.Integer, _  
    ByVal VerticalPlacement As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim EntitiesToDimension As System.Integer  
Dim HorizontalScheme As System.Integer  
Dim HorizontalPlacement As System.Integer  
Dim VerticalScheme As System.Integer  
Dim VerticalPlacement As System.Integer  
Dim value As System.Integer  
  
value = instance.AutoDimension(EntitiesToDimension, HorizontalScheme, HorizontalPlacement,  
    VerticalScheme, VerticalPlacement)
```

C#

```
System.int AutoDimension(  
    System.int EntitiesToDimension,  
    System.int HorizontalScheme,  
    System.int HorizontalPlacement,  
    System.int VerticalScheme,  
    System.int VerticalPlacement  
)
```

C++/CLI

```
System.int AutoDimension(  
& System.int EntitiesToDimension,  
& System.int HorizontalScheme,  
& System.int HorizontalPlacement,  
& System.int VerticalScheme,  
& System.int VerticalPlacement  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EntitiesToDimension

HorizontalScheme

HorizontalPlacement

VerticalScheme

VerticalPlacement

Visual Basic for Applications (VBA) Syntax

See [Sketch::AutoDimension](#).

See Also

[ISketch Interface](#)

[ISketch Members](#)

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AutoDimension2 Method (ISketch)

See Also [Example](#)



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Obsolete. Superseded by [ISketchManager::FullyDefineSketch](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AutoDimension2( _  
    ByVal EntitiesToDimension As System.Integer, _  
    ByVal HorizontalScheme As System.Integer, _  
    ByVal HorizontalPlacement As System.Integer, _  
    ByVal VerticalScheme As System.Integer, _  
    ByVal VerticalPlacement As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim EntitiesToDimension As System.Integer  
Dim HorizontalScheme As System.Integer  
Dim HorizontalPlacement As System.Integer  
Dim VerticalScheme As System.Integer  
Dim VerticalPlacement As System.Integer  
Dim value As System.Integer  
  
value = instance.AutoDimension2(EntitiesToDimension, HorizontalScheme, HorizontalPlacement,  
    VerticalScheme, VerticalPlacement)
```

C#

```
System.int AutoDimension2(  
    System.int EntitiesToDimension,  
    System.int HorizontalScheme,  
    System.int HorizontalPlacement,  
    System.int VerticalScheme,  
    System.int VerticalPlacement  
)
```

C++/CLI

```
System.int AutoDimension2(  
& System.int EntitiesToDimension,  
& System.int HorizontalScheme,  
& System.int HorizontalPlacement,
```

```
& System.int VerticalScheme,  
& System.int VerticalPlacement  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EntitiesToDimension

Entities to dimension as defined in [swAutodimEntities_e](#)

HorizontalScheme

Horizontal dimensioning scheme as defined in [swAutodimScheme_e](#)

HorizontalPlacement

Placement relative to the sketch as defined in [swAutodimHorizontalPlacement_e](#)

VerticalScheme

Vertical dimensioning scheme as defined in [swAutodimScheme_e](#)

VerticalPlacement

Placement relative to the sketch as defined in [swAutodimVerticalPlacement_e](#)

Return Value

swAutodimStatusSuccess if the sketch is automatically dimensioned successfully; see [swAutodimStatus_e](#) for values for reasons for possible failures

Visual Basic for Applications (VBA) Syntax

See [Sketch::AutoDimension2](#).

Example

[Autodimension a Sketch \(VBA\)](#)

[Insert DXF File and Add Dimension \(VBA\)](#)

[Autodimension All Sketches \(C#\)](#)

[Autodimension All Sketches \(VB.NET\)](#)

[Autodimension all Sketches \(VBA\)](#)

Remarks

If the EntitiesToDimension argument takes the value swAutodimEntitiesSelected, then use [IModelDocExtension::SelectByID2](#) with a mark value of swAutodimMarkEntities to select the sketch entities to dimension.

You can supply datums for horizontal and vertical dimensioning schemes. Select and mark a unique vertical edge, vertex, sketch point, or vertical sketch line as the datum for the horizontal dimensioning scheme, using swAutodimMarkHorizontalDatum as the mark value. Similarly, select a unique horizontal edge, vertex, sketch point, or horizontal sketch line as the datum for the vertical dimensioning scheme, using swAutodimMarkVerticalDatum as the mark value. It is an error to supply just one datum as indicated by the status value swAutodimStatus_DatumNotSupplied.

See Also

[ISketch Interface](#)

[ISketch Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CheckFeatureUse Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : CheckFeatureUse Method (ISketch)

Checks to see if this sketch is valid for use in creating a specified feature.

.NET Syntax

Visual Basic (Declaration)

```
Function CheckFeatureUse( _  
    ByVal FeatureType As System.Integer, _  
    ByRef OpenCount As System.Integer, _
```

```
ByRef ClosedCount As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim FeatureType As System.Integer  
Dim OpenCount As System.Integer  
Dim ClosedCount As System.Integer  
Dim value As System.Integer  
  
value = instance.CheckFeatureUse(FeatureType, OpenCount, ClosedCount)
```

C#

```
System.int CheckFeatureUse(  
    System.int FeatureType,  
    out System.int OpenCount,  
    out System.int ClosedCount  
)
```

C++/CLI

```
System.int CheckFeatureUse(  
    & System.int FeatureType,  
    & [Out] System.int OpenCount,  
    & [Out] System.int ClosedCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureType

Determine if this type of feature can be created as defined in [swSketchCheckFeatureProfileUsage_e](#)

OpenCount

Number of open contours found in this sketch

ClosedCount

Number of closed contours found in this sketch

Return Value

swSketchCheckFeatureStatus_OK if this sketch can be used to create the specified feature; see [swSketchCheckFeatureStatus_e](#) for possible failure values

Visual Basic for Applications (VBA) Syntax

See [Sketch::CheckFeatureUse](#).

Example

[Determine If Sketch Suitable for Feature \(VBA\)](#)

Remarks

This method is equivalent to the SOLIDWORKS interactive tool **Check Sketch for Feature Usage**. See the SOLIDWORKS Help for details.

The OpenCount and ClosedCount arguments are output values. If this method returns swSketchCheckFeatureStatus_OK, meaning that the sketch can be used to create the specified feature, then these two arguments contain useful information. If this method returns something else, then OpenCount and ClosedCount both return 0.

If the featureType value is not valid, this method returns swSketchCheckFeatureStatus_UnknownError.

See Also

[ISketch Interface](#)
[ISketch Members](#)

Availability

ConstrainAll Method (ISketch)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : ConstrainAll Method (ISketch)

Attempts to solve all of the apparent relations in the sketch and returns the number of constraints that were added to the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function ConstrainAll() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.ConstrainAll()
```

C#

```
System.int ConstrainAll()
```

C++/CLI

```
System.int ConstrainAll();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of constraints added

Visual Basic for Applications (VBA) Syntax

See [Sketch::ConstrainAll](#).

Remarks

If the sketch already has constraints, then no constraints are added (in the same way that the user-interface button **Constrain All** can only be pressed once for a sketch.)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetConstrainedStatus Method](#)

GetArcCount Method (ISketch)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetArcCount Method (ISketch)

Gets the number of arcs in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetArcCount()
```

C#

```
System.int GetArcCount()
```

C++/CLI

```
System.int GetArcCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of arcs in the sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetArcCount](#).

Example

[Get Arcs in Sketch \(VBA\)](#)
[Get Polyline Information \(VBA\)](#)

Remarks

Call this method before [ISketch::!GetArcs2](#) to determine the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetArcs2 Method](#)
[ISketchArc Interface](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetArcs Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetArcs Method (ISketch)

Obsolete. Superseded by [ISketch::GetArcs2](#) and [ISketch::!GetArcs2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcs() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetArcs()
```

C#

```
System.object GetArcs()
```

C++/CLI

```
System.Object^ GetArcs();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetArcs](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetArcs2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetArcs2 Method (ISketch)

Gets all of the arcs in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcs2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetArcs2()
```

C#

```
System.Object GetArcs2()
```

C++/CLI

```
System.Object^ GetArcs2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles containing the arcs in the sketch (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetArcs2](#).

Example

[Get Arcs in Sketch \(VBA\)](#)

Remarks

To determine the number of arcs, use [ISketch::GetArcCount](#).

Return value is an array of doubles with the format:

[*Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*], [*Center*] *Rotation Dir*]

where

- Color* - COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- Type* - Line type. Valid returns as defined in [swLineTypes_e](#). A line type is a combination of a line style and line weight.
- Line Font* - Line style. Valid line styles as defined in [swLineStyles_e](#) enumeration.
- Line Width* - Integer value defining the line width. Valid width values as defined in [swLineWeights_e](#).
- Layer ID* - Integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).

- *Layer Override* - Integer with bit flags set to determine which properties, if any, have been overridden with respect to the layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if Layer Override was returned as 3, the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
 - *Start[3]* - An array of 3 doubles (X,Y,Z) describing the start point.
 - *End[3]* - An array of 3 doubles (X,Y,Z) describing the end point. If the arc is closed, then End[3] is the same point as the Start.
 - *Center[3]* - An array of 3 doubles (X,Y,Z) describing the center point.
 - *Rotation Dir* - Rotational direction (CW = -1, CCW = 1)
- This set of data repeats for each arc in the sketch. The size of the array is (NumArcs * 16).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetArcs2 Method](#)
[ISketchArc Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAutomaticSolve Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetAutomaticSolve Method (ISketch)

Checks whether the computation to solve the sketch geometry of the part as modifications are automatically performed.

VB.NET Syntax

Visual Basic (Declaration)

```
Function GetAutomaticSolve() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Boolean

value = instance.GetAutomaticSolve()
```

C#

```
System.bool GetAutomaticSolve()
```

C++/CLI

```
System.bool GetAutomaticSolve();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if automatic solving is on, false if off

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetAutomaticSolve](#).

Remarks

This method can be controlled with [ISketch::SetAutomaticSolve](#).

See Also

GetConstrainedStatus Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetConstrainedStatus Method (ISketch)

Gets the current constrained status of the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetConstrainedStatus() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetConstrainedStatus()
```

C#

```
System.int GetConstrainedStatus()
```

C++/CLI

```
System.int GetConstrainedStatus();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Constrained status as defined in [swConstrainedStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetConstrainedStatus](#).

Example

[Get Sketch Constraints \(VBA\)](#)
[Autodimension All Sketches \(C#\)](#)
[Autodimension All Sketches \(VB.NET\)](#)
[Autodimension All Sketches \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::ConstrainAll Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

GetContourEdgeCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetContourEdgeCount Method (ISketch)

Gets the number of edges for this sketch contour.

.NET Syntax

Visual Basic (Declaration)

```
Function GetContourEdgeCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Integer

value = instance.GetContourEdgeCount()
```

C#

```
System.int GetContourEdgeCount()
```

C++/CLI

```
System.int GetContourEdgeCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of edges

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetContourEdgeCount](#).

Remarks

If the sketch contour is not closed or contains interior non-construction elements, this method returns 0.

Call this method before calling [ISketch::IGetContourEdges](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetContourEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetContourEdges Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetContourEdges Method (ISketch)

Gets the edges for a sketch that has one contour.

.NET Syntax

Visual Basic (Declaration)

```
Function GetContourEdges( _
    ByVal Xform As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim Xform As System.Object
Dim value As System.Object

value = instance.GetContourEdges(Xform)
```

C#

```
System.object GetContourEdges(  
    System.object Xform  
)
```

C++/CLI

```
System.Object^ GetContourEdges(  
& System.Object^ Xform  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xform
Array of size 16 doubles representing the transform

Return Value

Array of [edges](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetContourEdges](#).

Remarks

This method supports sketches with one contour only. For sketches with multiple contours, use [ISketch::GetSketchContours](#) or [ISketch::IGetSketchContours](#) and then [ISketchContour::GetEdges](#) or [ISketchContour::IGetEdges](#).

Specifying the unit transform returns the edges in the space of the sketch.

If your application is expecting the returned edges to be in the context of the model's coordinate system, then xform should be the inverse of the transform returned by [ISketch::ModelToSketchTransform](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetContourEdgeCount Method](#)
[ISketch::IGetContourEdges Method](#)
[ISketchContour Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDetachSegmentOnDrag Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetDetachSegmentOnDrag Method (ISketch)

Gets the **Detach Segment on Drag** setting.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDetachSegmentOnDrag() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Boolean  
  
value = instance.GetDetachSegmentOnDrag()
```

C#

```
System.bool GetDetachSegmentOnDrag()
```

C++/CLI

```
System.bool GetDetachSegmentOnDrag();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if **Detach Segment on Drag** is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetDetachSegmentOnDrag](#).

Remarks

Although this setting can be returned or set at any time, SOLIDWORKS sets it to false upon exiting or entering a sketch. Thus, this method is only useful if called while a sketch is active.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::SetDetachSegmentOnDrag Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEllipseCount Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetEllipseCount Method (ISketch)

Gets the number of ellipses in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipseCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetEllipseCount()
```

C#

```
System.int GetEllipseCount()
```

C++/CLI

```
System.int GetEllipseCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of ellipses in the sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetEllipseCount](#).

Example

[Get Polylines Information \(VBA\)](#).

Remarks

Call this method before calling [ISketch::GetEllipses3](#) to determine the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetEllipses3 Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetEllipses Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetEllipses Method (ISketch)

Obsolete. Superseded by [ISketch::GetEllipses3](#) and [ISketch::IGetEllipses3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipses() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetEllipses()
```

C#

```
System.object GetEllipses()
```

C++/CLI

```
System.Object^ GetEllipses();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetEllipses](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetEllipses2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetEllipses2 Method (ISketch)

Obsolete. Superseded by [ISketch::GetEllipses3](#) and [ISketch::IGetEllipses3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipses2() As System.Object
```

```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetEllipses2()
```

C#

```
System.object GetEllipses2()
```

C++/CLI

```
System.Object^ GetEllipses2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetEllipses2](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEllipses3 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetEllipses3 Method (ISketch)

Gets all of the ellipses in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipses3() As System.Object
```

Visual Basic (Usage)

```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetEllipses3()
```

C#

```
System.object GetEllipses3()
```

C++/CLI

```
System.Object^ GetEllipses3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetEllipses3](#).

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchEllipse](#) objects.

The return values are in an array of doubles:

[Color, LineType, LineFont, LineWidth, LayerID, LayerOverride, StartPt[3], EndPt[3], CenterPt[3], MajorPt[3], MinorPt[3],Direction ...]

where:

Color :	COLORREF returned as an integer. Return value could be 0 or -1 for default color.
LineType :	Line type. Valid returns are defined in swLineTypes_e . A line type is a combination of a lineStyle and lineWeight.
LineFont :	Value is used for polyline font information. This value can be used as an input to the IDrawingDoc::GetLineFontInfo2 and IDrawinDoc::GetLineFontName2 .
LineWidth :	Integer value defining the line width. Valid width values as defined in swLineWeights_e .
LayerID:	Integer value indicating which layer holds this entity. Obtain the ILayer object by passing this integer value to ILayerMgr::GetLayerById or ILayerMgr::!GetLayerById .
LayerOverride:	Integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
StartPt[3] :	Array of 3 doubles (X,Y,Z) describing the ellipse start point
EndPt[3] :	Array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the StartPt.
CenterPt[3] :	Array of 3 doubles (X,Y,Z) describing the ellipse center point.
MajorPt[3] :	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the major axis.
MinorPt[3] :	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the minor axis.
Direction :	-1 for clockwise, +1 for counterclockwise

This set of data repeats for each ellipse in the sketch. The size of the array is (NumEllipses * 20).

To determine the number of ellipses, use [ISketch::GetEllipseCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::!GetEllipses3 Method](#)
[ISketchEllipse Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLineCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetLineCount Method (ISketch)

Obsolete. Superseded by [ISketch::GetLineCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetLineCount()
```

C#

```
System.int GetLineCount()
```

C++/CLI

```
System.int GetLineCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetLineCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLineCount2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetLineCount2 Method (ISketch)

Gets the number of lines in the sketch with an option to exclude or include crosshatch lines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineCount2( _  
    ByVal CrossHatchOption As System.Short _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim CrossHatchOption As System.Short  
Dim value As System.Integer  
  
value = instance.GetLineCount2(CrossHatchOption)
```

C#

```
System.int GetLineCount2(  
    System.short CrossHatchOption  
)
```

C++/CLI

```
System.int GetLineCount2(  
& System.short CrossHatchOption  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

Crosshatch option as defined in [swCrossHatchFilter](#) e

Return Value

Number of lines

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetLineCount2](#).

Example

[Get Lines in Sketch \(VBA\)](#)
[Get Polyline Information \(VBA\)](#)

Remarks

Call this method before calling [ISketch::IGetLines2](#) to determine the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLines Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetLines Method (ISketch)

Obsolete. Superseded by [ISketch::GetLines2](#) and [ISketch::IGetLines2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLines() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetLines()
```

C#

```
System.object GetLines()
```

C++/CLI

```
System.Object^ GetLines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetLines](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLines2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetLines2 Method (ISketch)

Gets all of the lines in the sketch with an option to include or exclude crosshatch lines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLines2( _
    ByVal CrossHatchOption As System.Short _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim CrossHatchOption As System.Short
Dim value As System.Object

value = instance.GetLines2(CrossHatchOption)
```

C#

```
System.object GetLines2(
    System.short CrossHatchOption
)
```

C++/CLI

```
System.Object^ GetLines2(
    & System.short CrossHatchOption
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

Crosshatch option as defined in [swCrossHatchFilter_e](#)

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetLines2](#).

Example

[Get Lines in Sketch \(VBA\)](#)

Remarks

The return value is the following array of doubles:

[*Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*]]

where all data values are returned as doubles:

Color = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

Type = line type. Valid returns as defined in [swLineTypes_e](#). A line type is a combination of a line style and line weight.

Line Font = line style. Valid line styles as defined in [swLineStyles_e](#).

Line Width = integer value defining the line width. Valid width values as defined in [swLineWeights_e](#).

Layer ID = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).

Layer Override = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride is returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

Start[3] = array of 3 doubles (X,Y,Z) describing the line start point.

End[3] = array of 3 doubles (X,Y,Z) describing the line end point.

This set of data repeats for each line in the sketch. The number of doubles returned is (lineCount * 12). To determine the number of lines in the sketch, use [ISketch::GetLineCount2](#).

See [ISketch::GetSketchSegments](#) or [ISketch::!EnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchLine](#) objects.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetLines2 Method](#)
[ISketchLine Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetParabolaCount Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetParabolaCount Method (ISketch)

Gets the number of parabolas in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetParabolaCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetParabolaCount()
```

C#

```
System.int GetParabolaCount()
```

C++/CLI

```
System.int GetParabolaCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of parabolas in the sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetParabolaCount](#).

Example

[Get Parabolas in Sketch \(VBA\)](#)
[Get Polyline Information \(VBA\)](#)

Remarks

Call this method before calling [ISketch::IGetParabolas2](#) to determine the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetParabolas2 Method](#)
[ISketchParabola Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetParabolas Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetParabolas Method (ISketch)

Obsolete. Superseded by [ISketch::GetParabolas2](#) and [ISketch::IGetParabolas2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetParabolas() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetParabolas()
```

C#

```
System.object GetParabolas()
```

C++/CLI

```
System.Object^ GetParabolas();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetParabolas](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetParabolas2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetParabolas2 Method (ISketch)

Gets all of the parabolas in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetParabolas2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetParabolas2()
```

C#

```
System.object GetParabolas2()
```

C++/CLI

```
System.Object^ GetParabolas2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetParabolas2](#).

Example

[Get Parabolas in Sketch \(VBA\)](#)

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchParabola](#) objects.

The return values are in an array of doubles:

[*Color*, *LineType*, *LineStyleIndex*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *FocusPt*[3], *ApexPt*[3]]

where:

Color = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

LineType = line type. Valid returns as defined in [swLineTypes_e](#). A line type is a combination of a line style and line weight.

LineStyleIndex = line style. Valid line styles as defined in [swLineStyles_e](#).

LineWidth = integer value defining the line width. Valid width values as defined in [swLineWeights_e](#).

LayerID = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).

LayerOverride = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

StartPt[3] = array of 3 doubles (X,Y,Z) describing the parabola start point.

EndPt[3] = array of 3 doubles (X,Y,Z) describing the parabola end point.

FocusPt[3] = array of 3 doubles (X,Y,Z) describing the parabola focus point.

ApexPt[3] = array of 3 doubles (X,Y,Z) describing the parabola apex point.

This set of data repeats for each parabola in the sketch. The size of the array is (NumParabolas * 18). To determine the number of parabolas, see [ISketch::GetParabolaCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetParabolas2 Method](#)
[ISketchParabola Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPolyLineCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetPolyLineCount Method (ISketch)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolyLineCount( _  
    ByRef PointCount As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim PointCount As System.Integer
Dim value As System.Integer

value = instance.GetPolyLineCount(PointCount)
```

C#

```
System.int GetPolyLineCount(
    ref System.int PointCount
)
```

C++/CLI

```
System.int GetPolyLineCount(
    & System.int% PointCount
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPolylines Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetPolylines Method (ISketch)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolylines() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetPolylines()
```

C#

```
System.object GetPolylines()
```

C++/CLI

```
System.Object^ GetPolylines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetReferenceEntity Method (ISketch)

[See Also](#) [Example](#)



Gets the entity on which this sketch was created.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetReferenceEntity( _  
    ByRef LEntityType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim LEntityType As System.Integer  
Dim value As System.Object  
  
value = instance.GetReferenceEntity(LEntityType)
```

C#

```
System.object GetReferenceEntity(  
    out System.int LEntityType  
)
```

C++/CLI

```
System.Object^ GetReferenceEntity(  
& [Out] System.int LEntityType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LEntityType

Entity type as defined in [swSelectType_e](#) (only swSelfFACES and swSelDATUMPLANES are valid values for this method)

Return Value

Either a [reference plane](#) or a [face](#), depending on the value of IEntityType

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetReferenceEntity](#).

Example

[Get Plane On Which Sketch Created \(VBA\)](#)
[Get Plane or Face for Sketch \(VBA\)](#)

Remarks

If the sketch resides on a face that is consumed by subsequent features, then this method returns NULL and swSelNOTHING. To get to the face, edit the sketch using [IModelDoc2::EditSketch](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchBlockInstanceCount Method (ISketch)

Gets the number of block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchBlockInstanceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Integer

value = instance.GetSketchBlockInstanceCount()
```

C#

```
System.int GetSketchBlockInstanceCount()
```

C++/CLI

```
System.int GetSketchBlockInstanceCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of block instances in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchBlockInstanceCount](#).

Remarks

Call this method before calling [ISketch::IGetSketchBlockInstances](#) to get the size of the array for that method.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchBlockInstances Method](#)
[ISketchBlockInstance Interface](#)
[ISketchBlockDefinition Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchBlockInstances Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchBlockInstances Method (ISketch)

Gets the block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchBlockInstances() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchBlockInstances()
```

C#

```
System.object GetSketchBlockInstances()
```

C++/CLI

```
System.Object^ GetSketchBlockInstances();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the [block instances](#) in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchBlockInstances](#).

Remarks

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchBlockInstanceCount Method](#)
[ISketch::IGetSketchBlockInstances Method](#)
[ISketchBlockInstance Interface](#)
[ISketchBlockDefinition Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchContourCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchContourCount Method (ISketch)

Gets the number of sketch contours in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchContourCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetSketchContourCount()
```

C#

```
System.int GetSketchContourCount()
```

C++/CLI

```
System.int GetSketchContourCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of sketch contours

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchContourCount](#).

Remarks

Call this method before calling [ISketch::IGetSketchContours](#) to get the size of the array for that method.

This method works even if the sketch is suppressed.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchContours Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSketchContours Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchContours Method (ISketch)

Gets the sketch contours in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchContours() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchContours()
```

C#

```
System.object GetSketchContours()
```

C++/CLI

```
System.Object^ GetSketchContours();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch contours](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchContours](#).

Example

[Get Sketch Contours \(VBA\)](#)
[Get Corresponding Objects in Assembly Component \(C#\)](#)
[Get Corresponding Objects in Assembly Component \(VB.NET\)](#)
[Get Corresponding Objects in Assembly Component \(VBA\)](#)

Remarks

This method works even if the sketch is suppressed.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchContourCount Method](#)
[ISketch::IGetSketchContours Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchHatches Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchHatches Method (ISketch)

Gets an array of sketch hatches that exist in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchHatches() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchHatches()
```

C#

```
System.object GetSketchHatches()
```

C++/CLI

```
System.Object^ GetSketchHatches();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch hatches](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchHatches](#).

Example

[Get Area Hatch \(VBA\)](#)
[Get Hatching Data \(VBA\)](#)
[Insert Hatch \(VBA\)](#)

Remarks

For information about hatches:

- in drawing views, use [View::GetFaceHatchCount](#), [View::GetFaceHatches](#), and [View::IGetFaceHatches](#).
- detail, broken, or crop views, use [View::GetSolidHatchCount](#), [View::GetSolidHatchInfo](#), and [View::IGetSolidHatchInfo](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IEnumSketchHatches Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchPathCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPathCount Method (ISketch)

Gets the number of sketch paths in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchPathCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetSketchPathCount()
```

C#

```
System.int GetSketchPathCount()
```

C++/CLI

```
System.int GetSketchPathCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of sketch paths

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPathCount](#).

Remarks

Call this method before calling [ISketch::IGetSketchPaths](#) to get the sketch paths in this sketch.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPaths Method](#)
[ISketchPath Interface](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchPaths Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPaths Method (ISketch)

Gets the sketch paths in this sketch.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetSketchPaths() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetSketchPaths()
```

C#

```
System.object GetSketchPaths()
```

C++/CLI

```
System.Object^ GetSketchPaths();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch paths](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPaths](#).

Example

[Create Sketch Path \(VBA\)](#)
[Create Sketch Path \(VB.NET\)](#)
[Create Sketch Path \(C#\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPathCount Method](#)
[ISketch::IGetSketchPaths Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSketchPictureCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPictureCount Method (ISketch)

Gets the number of pictures on this sketch.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetSketchPictureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Integer

value = instance.GetSketchPictureCount()
```

C#

```
System.int GetSketchPictureCount()
```

C++/CLI

```
System.int GetSketchPictureCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of pictures on this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPictureCount](#).

Remarks

Call this method before calling [ISketch::IGetSketchPictures](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPictures Method](#)
[ISketchPicture Interface](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchPictures Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPictures Method (ISketch)

Gets the pictures on this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchPictures() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchPictures()
```

C#

```
System.object GetSketchPictures()
```

C++/CLI

```
System.Object^ GetSketchPictures();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [pictures](#) on this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPictures](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPictureCount Method](#)
[ISketch::!GetSketchPictures Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchPoints Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPoints Method (ISketch)

Obsolete. Superseded by [ISketch::GetSketchPoints2](#), [ISketch::!GetSketchPoints2](#), and [ISketch::!EnumSketchPoints](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchPoints() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchPoints()
```

C#

```
System.object GetSketchPoints()
```

C++/CLI

```
System.Object^ GetSketchPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPoints](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchPoints2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPoints2 Method (ISketch)

Gets the sketch points in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchPoints2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object
```

```
value = instance.GetSketchPoints2()
```

C#

```
System.Object GetSketchPoints2()
```

C++/CLI

```
System.Object^ GetSketchPoints2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch points](#) in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPoints2](#).

Example

[Delete All Constraints in Selected Sketch \(VBA\)](#)
[Get Sketch Points and Their Persistent IDs \(VBA\)](#)
[Get Sketch Points \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Insert DXF File and Add Dimension \(VBA\)](#)
[Transform Sketch to Model \(VBA\)](#)
[Autodimension All Sketches \(C#\)](#)
[Autodimension All Sketches \(VB.NET\)](#)
[Autodimension All Sketches \(VBA\)](#)

Remarks

The difference between this method and the now obsolete method [ISketch::GetSketchPoints](#) is that this method does not return sketch points intended for internal SOLIDWORKS use only.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetSketchPoints2 Method](#)
[ISketch::IEnumSketchPoints Method](#)
[ISketch::GetUserPointsCount Method](#)
[ISketch::GetUserPoints2 Method](#)
[ISketch::IGetUserPoints2 Method](#)
[ISketchPoint Interface](#)
[ISketch::MergePoints Method](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchPointsCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPointsCount Method (ISketch)

Obsolete. Superseded by [ISketch::GetSketchPointsCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchPointsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer
```

```
value = instance.GetSketchPointsCount()
```

C#

```
System.int GetSketchPointsCount()
```

C++/CLI

```
System.int GetSketchPointsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPointsCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSketchPointsCount2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchPointsCount2 Method (ISketch)

Gets the number of sketch points in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchPointsCount2() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer
```

```
value = instance.GetSketchPointsCount2()
```

C#

```
System.int GetSketchPointsCount2()
```

C++/CLI

```
System.int GetSketchPointsCount2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of [sketch points](#) in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchPointsCount2](#).

Example

[Get Spline Points \(C++\)](#)

Remarks

Call this method before calling [ISketch::IGetSketchPoints2](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPoints2 Method](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

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GetSketchRegionCount Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchRegionCount Method (ISketch)

Gets the number of sketch regions in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchRegionCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetSketchRegionCount()
```

C#

```
System.int GetSketchRegionCount()
```

C++/CLI

```
System.int GetSketchRegionCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of sketch regions

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchRegionCount](#).

Example

[Get Sketch Regions \(VBA\)](#)
[Get Sketch Regions \(VB.NET\)](#)
[Get Sketch Regions \(C#\)](#)

Remarks

Call this method before calling [ISketch::IGetSketchRegions](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketchRegion::GetSketchRegions Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

GetSketchRegions Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchRegions Method (ISketch)

Gets the sketch regions in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchRegions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchRegions()
```

C#

```
System.object GetSketchRegions()
```

C++/CLI

```
System.Object^ GetSketchRegions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch regions](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchRegions](#).

Example

[Get Sketch Regions \(VBA\)](#)
[Get Sketch Regions \(VB.NET\)](#)
[Get Sketch Regions \(C#\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchRegionCount Method](#)
[ISketch::IGetSketchRegions Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

GetSketchSegments Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchSegments Method (ISketch)

Gets the sketch segments in this sketch, which include line, arc, spline, parabola, and ellipse entities.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchSegments() As System.Object
```

Visual Basic (Usage)

```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchSegments()
```

C#

```
System.Object GetSketchSegments()
```

C++/CLI

```
System.Object^ GetSketchSegments();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch segments](#), which include line, arc, spline, parabola, and ellipse entities, in this sketch (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchSegments](#).

Example

[Delete All Constraints in Selected Sketch \(VBA\)](#)
[Find Total Length of Sketch Segments in Sketch \(VBA\)](#)
[Get All Elements of Sketch \(VBA\)](#)
[Get All Sketch Segments in Drawing Template \(VBA\)](#)
[Get Sketch Constraints \(VBA\)](#)
[Insert DXF File and Add Dimension \(VBA\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(C#\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(VB.NET\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(VBA\)](#)

Remarks

If the sketch has errors, this method does not return all of the sketch segments.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IEnumSketchSegments Method](#)
[ISketch::GetSketchTextSegments Method](#)
[ISketch::IEnumSketchTextSegments Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

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GetSketchSlotCount Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchSlotCount Method (ISketch)

Gets the number of sketch slots in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchSlotCount() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketch
Dim value As System.Integer

value = instance.GetSketchSlotCount()
```

C#

```
System.int GetSketchSlotCount()
```

C++/CLI

```
System.int GetSketchSlotCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of sketch slots in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchSlotCount](#).

Example

[Get Sketch Slot Data \(VBA\)](#)

Remarks

Call this method before calling [ISketch::IGetSketchSlots](#) to determine the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchSlots Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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GetSketchSlots Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchSlots Method (ISketch)

Gets the sketch slots in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchSlots() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetSketchSlots()
```

C#

```
System.object GetSketchSlots()
```

C++/CLI

```
System.Object^ GetSketchSlots();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch slots](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchSlots](#).

Example

[Get Sketch Slot Data \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetSketchSlots Method](#)
[ISketch::GetSketchSlotCount Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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GetSketchTextSegments Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSketchTextSegments Method (ISketch)

Gets the sketch segments that represent the selected text in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchTextSegments() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSketchTextSegments()
```

C#

```
System.object GetSketchTextSegments()
```

C++/CLI

```
System.Object^ GetSketchTextSegments();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch segments](#) for the selected text in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSketchTextSegments](#).

Example

[Get Sketch Entities \(VBA\)](#)
[Replace Sketch Text \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IEnumSketchTextSegments Method](#)
[ISketch::GetSketchSegments Method](#)
[ISketch::IEnumSketchSegments Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0